Title: **Electrical Safety Policy**
Document Type: **Work Instruction**
Document Number: **SGOP-0022** **Revision Number:** **23**

1.0 PURPOSE

1.1 The purpose of this policy is to provide a safe working area for employees relative to the hazards arising from the use of electricity. Specific practices are to be consistent with the degree of hazard present including, but not limited to, electric shock, falls due to shock, thermal burns, arc flash burns, and arc blast.

1.2 This policy addresses electrical safety related work practices, safety-related maintenance requirements, and other administrative controls for employee workplaces that are necessary for the safeguarding of employees relative to the hazards associated with electrical energy during activities such as the installation, inspection, operation, maintenance, and demolition of electric conductors, electric equipment, signaling and communications conductors and equipment, and raceways. This procedure also includes safe work practices for employees performing other work activities that can expose them to electrical hazards.

1.3 If a situation arises that is not specifically covered under this policy, consult with a Blue Card holder prior to designing or performing work on electrical equipment.

1.4 This policy is applicable to all the organizations at Johnson & Johnson Vision Care (JJVC) Johnson and Johnson Vision Care, Inc. located in Jacksonville, Florida.

1.5 This policy applies to all work performed.

1.5.1 on or near un-insulated, un-guarded, energized conductors or circuit parts that involves activities inside the Limited Approach Boundary and/or the Arc Flash Boundary, or

1.5.2 involving an interaction with equipment where conductors or circuit parts are not exposed, but an increased risk of injury from an exposure to arc flash exists.

1.5.3 References:
1.5.3.1 OSHA 29 CFR 1910, Subpart S
1.5.3.2 NFPA 70E-2021, Standard for Electrical Safety in the Workplace
1.5.3.3 NFPA 79-2021, Electrical Standard for Industrial Machinery
1.5.3.4 EN 50110-1, “Operation of electrical installations.”
1.5.3.5 IEC 60204-1, “Electrical equipment of machinery.”
1.5.3.6 TSG_22_3_001, “J&J Electrical Systems Safety Program (ESSP)

2.0 Definitions

2.1 Arc Flash Hazard: A dangerous source of possible injury or damage to health associated with the released of energy caused by an electric arc.

2.2 Arc Flash PPE Category (AF PPE): A method used to represent the level of arc flash hazard present at a working distance of 18 inches, should an arc flash occur by grouping ranges of predicted arc flash incident energy into categories that is then used to determine the appropriate arc rated PPE to be worn. Arc Flash PPE Categories for the site are listed below. Category 0 is a JJVC requirement.

2.3

	Arc Flash PPE Category	Arc flash energy range (cal/cm^2)
2.3.1	0	0 to 1.2
2.3.2	1	> 1.2 to 4
2.3.3	2	> 4 to 8
2.3.4	3	> 8 to 25
2.3.5	4	> 25 to 40

NOTE: Electrical work that could expose an individual to arc flash energy above 40 cal/cm^2 is prohibited from being performed while energized.

2.4 Permanently Mounted Absence of Voltage Tester: An adequately rated, permanently mounted device used to test for absence of voltage during the process of Establishing an Electrically Safe Work Condition. The device must be 1) installed in accordance with manufacturer's instructions and tests the conductors and circuit parts at the point of work, 2) listed and labeled for the purpose of testing for absence of voltage, 3) capable of testing each phase conductor phase-to-phase and phase-to-ground and 4) capable of verifying satisfactory operation on a known voltage source before and after testing for absence of voltage.

2.5 Arc Rating: The value attributed to materials that describes their performance to exposure to an electrical arc discharge. The arc rating is expressed in cal/cm^2 and is derived from the determined value of the arc thermal performance value (ATPV) or energy of break open threshold (E_{BT}) (should a material system exhibit a break open response below the ATPV value).

2.6 Attendant: A person who has been made aware of the boundaries and hazards, with the primary responsibility of warning and keeping persons out of the work area other than those performing the work. The attendant shall remain stationed as long as there is the potential for employees to be exposed to electrical hazards.

2.7 Balaclava (Sock Hood): An arc-rated head covering that protects the neck and head except for the facial area of the eyes and nose.

2.8 Boundary, Arc Flash: An arc flash hazard approach limit at a distance from a prospective arc source within which a person could receive the onset of a non-curable, second-degree burn (1.2 cal/cm^2) if an electrical arc flash were to occur. Arc Rated PPE is required in this area.

2.9 Boundary, Limited Approach: An electrical shock hazard approach limit at a distance from an exposed energized
part within which a shock hazard exists, and only Qualified Persons are allowed, or Unqualified Persons only when continuously escorted by a Blue Card Person. Electric shock and arc rated PPE is required in this area.

2.10 Boundary, Restricted Approach: An electrical shock hazard approach limit at a distance from an exposed energized
part within which there is an increased risk of shock due to electrical arc over combined with inadvertent movement for personnel working in close proximity to the energized part. Only Qualified individuals are allowed in this area. Electric shock and arc rated PPE is required in this area.

2.11 Conductive Materials: Materials capable of carrying an electrical current, including insulated materials rated less than the working voltage.

2.12 Electrically Operated Cord and Plug Connected Equipment: Electrically operated equipment that is powered through a receptacle and plug such as portable equipment and equipment designed to be taken in and out of service (more than 12 times a year).

NOTE: Stationary plug and cord connected equipment (such as personal computers, water coolers, emergency lights, household appliances, and other such machinery), that are installed such that the connecting cord and plug are not subject to physical damage during normal use, need not be inspected or tested except when they are initially installed, relocated or repaired. Reference VWA-0837.

2.13 Daisy Chaining: The connecting together of more than one multi-outlet power strips and cords to form a continuous link for the purpose of increasing the number of available outlets and/or the available length of cord.

2.14 Diagnostic Work: Routine tasks performed utilizing approved testing equipment and diagnostic work procedures by Qualified Persons.

2.15 Electrical Hazard: A dangerous condition such that contact or equipment failure can result in electric shock, arc flash burn, thermal burn, or arc blast injury.

2.16 Electrically Safe Work Condition: A state in which an electrical conductor or circuit part to be worked on or near has been disconnected from energized parts, locked/tagged in accordance with the HECP (**SGOP-0028**), tested to ensure the absence of voltage and grounded if determined necessary.

2.17 Energized Components: Electrical equipment or circuits that have not been placed in an electrically safe work condition.

2.18 Energized Electrical Diagnostic Work (Diagnostic Work): Intentionally coming in contact with energized electrical conductors or circuit parts with probes or test equipment, regardless of the personal protective equipment (PPE) a person is wearing. Any work that does not require making any physical change to the equipment, such as taking voltage or current measurements (using a clamp on meter) on electrical equipment with approved test equipment.

2.19 Energized Electrical Repair Work (Repair Work): Intentionally coming in contact with energized electrical conductors or circuit parts (50 volts AC/DC and above) with the hands, feet, or other body parts, or tools, regardless of the personal protective equipment (PPE) a person is wearing. Examples would include making, tightening or breaking connections, removing or replacing components, work on de-energized conductors adjacent to exposed energized circuit parts such that people are within the arc flash and/or shock boundaries, etc. This type of work requires an Live Electrical Work PTW in the ePTW system (**VCF-2335** is the backup form that may be used if the ePTW system is down, with EH&S Approval)

NOTE: There are two categories of "energized work": *Diagnostic Work* is taking readings or measurements of electrical equipment with approved test equipment that does not require making any physical change to the equipment; *Repair Work* is any physical alteration of electrical equipment (such as making or tightening connections, removing or replacing components, etc.).

2.20 Energy Isolating Device: A mechanical device that physically prevents the transmission or release of energy such as manually operated electrical circuit breakers; disconnect switches and line valves or blocks.

2.21 Energy Source: The root of any chemical, electrical, thermal, pneumatic, mechanical or hydraulic energy (energy can exist in an active or stored state).

2.22 Exposed (as applied to energized electrical conductors or circuit (energized) parts): Capable of being inadvertently.
touched or approached nearer than a safe distance by a person. It is applied to electrical conductors or circuit parts that are not suitably guarded, isolated, or insulated, or are not finger touch protected, not IP 20 rated.

2.23 Finger Touch Protection: A means of preventing an accidental contact with energized parts.

NOTE: The preferred installation would meet the IEC-60529 IP-20 or greater requirement. An acceptable alternative is to provide a barrier with a "Lexan®" guard. For diagnostic work, the openings in the guard should meet the "Zero Access" minimum distance to the energized part.

2.24 Fixed Electrical Equipment: Electrical equipment fastened in place or connected by permanent wiring methods.

2.25 Flammable Materials: As applied to this policy, flammable materials include, but are not limited to:

2.25.1 Flammable gases, vapors or liquids.
2.25.2 Combustible dust.
2.25.3 Ignitable fibers of airborne particles.

2.26 Ground-Fault Circuit Interrupter (GFCI): A device intended for the protection of personnel that functions to de-energize a circuit or portion thereof within an established period of time when a current to ground is between 4 mA and 6 mA.

2.27 Grounding Conductor: A conductor used to connect equipment or the grounded circuit of a wiring system to a grounding electrode or electrodes, and that is not intended to carry current under normal conditions.

2.28 Hazard: A source of possible injury or damage to health.

2.29 Hazardous: Involving exposure to at least one hazard.

2.30 IEC: International Electrotechnical Commission

2.31 Incident Energy: The amount of thermal energy impressed on a surface, a certain distance from the source, generated during an electrical arc event. Incident energy is typically expressed in calories per square centimeter (cal/cm^2).

2.32 Incident Energy Analysis: A component of an arc flash risk assessment used to predict the incident energy of an arc flash for a specified set of conditions.

2.33 Intrinsically Safe (IS): Equipment or devices approved for use in areas of an explosive nature such as where solvents and/or dust is present.

2.34 Isolate: Complete disconnection of a device from other devices and circuits.

2.35 Insulating Materials: Materials not capable of carrying an electrical current.

2.36 JJVC Electrical Board (VEB): A group of senior electrical engineers and electricians identified and selected by JJVC EH&S. (Refer to the "Responsibility section" below). This team is considered the Local Electrical Safety. Work Practices Team. Members shall be educated in current local standards and NFPA 70E.

2.37 Job Hazard Analysis (Risk Assessment): An overall process that identifies hazards, estimates the potential severity of injury or damage to health, estimates the likelihood of occurrence of injury or damage to health, and determines if protective measures are required.

NOTE: As used in this policy, arc flash risk assessment and shock risk assessment are types of risk assessments.

2.38 Left (off) Hand Rule: A work procedure for safety operating an electrical disconnecting device such as a circuit breaker or disconnect switch. Procedure steps are as follows:

2.38.1 Stand to the side of the disconnecting device so you are not in front of the enclosure.
2.38.2 Grasp the disconnecting device operating handle with your LEFT hand if facing the disconnecting device, RIGHT hand if facing away from the disconnecting device.
2.38.3 Turn your body and face away from the disconnecting device.
2.38.4 Close your eyes.
2.38.5 Take a deep breath and hold it.
2.38.6 Move the disconnecting device operating handle to the ON or OFF position.

2.39 Energized Electrical Working Spaces/Areas: A space around electrically energized components in which an isolation level to prevent danger is not assured when encroaching it without protective measures.

2.40 Lockout: The placement of a lockout device and warning tag on an energy isolating device in accordance with equipment and installation specific procedures, ensuring that the equipment it controls cannot be operated until the locking device is removed. Under no circumstance is a tag to be used without a lockout device.

2.40a. Out of Service Tag:
The warning tag must communicate the identity of the person who is responsible for the placement of the lock and tag, and the reason for the application of the Lockout (e.g. machine under repair and part removed, part defective and needs repair, equipment preventive maintenance).

2.41 Lockout Device: A device that utilizes a positive means (ex. a key lock) to secure an energy isolating device in a safe position. The locks used must be unique and consistent throughout the facility and used for lockout purposes only. Reference **SGOP-0028**.

2.42 Metrix MX 57EX: An approved Intrinsically Safe (IS) handheld electrical test instrument

2.43 NEC®: National Electric Code

2.44 NFPA: National Fire Protection Association

2.45 Neutral Conductor: The conductor connected to the neutral point of a power system that is intended to carry current under normal conditions.

2.46 Nominal Voltage: Ranges of circuit voltages established at the site for the purpose of assigning work to individuals of differing qualifications, knowledge and skills, defined as follows:

2.46.1	Extra Low Voltage:	Not exceeding 50 volts AC/DC
2.46.2	Low Voltage:	Between 50 – 600 volts AC/DC
2.46.3	High Voltage:	Exceeding 600 volts AC/DC

2.47 Qualified Person: One who has demonstrated skills and knowledge related to the construction and operation of the electrical equipment and installation and has received safety training to recognize and avoid the hazards involved. These individuals have been trained and are competent at avoiding the hazards of working on or near electrically energized components and are authorized by management to perform such work.

NOTE: There are two classifications of JJVC® employees that are considered Qualified Persons: Green Card holder and Blue Card holder. This SGOP documents the certification process and the type of work allowed for each of the classifications.

2.48 Risk: A combination of the likelihood of occurrence of injury or damage to health and the severity of injury or damage to health that results from a hazard.

2.49 Special Operating Conditions (from an electrical perspective): Areas, installation methods, equipment, work practices, etc. unique to the site that have the potential to present safety hazards to personnel, that would not be encountered at other similar facilities.

2.50 Strain Relief: Devices that are required to protect an internal cord and conductor from damage due to flexing at the exit point from the rigid hardware or electrical fitting on the equipment.

2.51 FLUKE T5: A brand of handheld electrical test instrument

2.52 Unqualified Persons: Those individuals who may reasonably be expected to face comparable risk of injury due to electric shock or other electrical hazards that have little or no training on avoiding the hazards of working on, or near energized electrical components. This includes, but is not limited to, individuals who work on or about machinery, equipment or power distribution devices conducting their normal workplace activities (e.g., mechanics, painters, engineers, research scientist/technicians, facility technicians and others who may participate in the maintenance, repair and diagnostic work of certain types of machinery/equipment as part of their assigned responsibility).

2.53 VOM: Volt-Ohm meter

2.54 Work in Progress Tag: A tag that must be attached to exposed wiring that is not terminated but is in a safe condition that identifies the project name, contractor company and/or the JJVC person responsible for that project.

2.55 Working On energized parts 50 Volts AC/DC or higher: Coming in direct contact with energized electrical conductors/components with hands, feet, or other body parts while working with tools, probes, test equipment, etc., regardless of the personal protective equipment a person may be wearing.

NOTE 1: This type of work is not allowed under this SGOP other than diagnostic work and routine operation work mentioned below. No JJVC Employee shall perform any other type of energized work. If any other energized work is needed, only approved contractors shall perform the work, and only after an Live Electrical Work Permit in the ePTW System, has been issued and approved.

NOTE 2: Shock hazard is significantly reduced if the circuit voltage is less than 50Volts AC/DC– caution should still be exercised because under certain conditions, such as wet locations, circuits operating under 50 volts AC/DC can pose a shock hazard. All such instances should be evaluated by an electrically certified individual.

2.56 Working Near (energized parts): Any activity inside the Limited Approach Boundary.

3.0 Electrical Certification Requirements

3.1 General

3.1.1 An Electrical Certification Program exists at the site that establishes minimum skills, knowledge and ability criteria for JJVC® employees and directly supervised temporary employees to be considered Qualified to perform certain tasks. The Electrical Certification Program identifies “Green Card” and “Blue Card” holder classifications, and the requirements for each to be authorized to perform certain electrical work. The following are the requirements for the Electrical Certification program.

3.1.2 The electrical certification program will be administered by a VEB board member. Only VEB Board approved individuals shall conduct the classroom and hands-on electrical certification training.

3.1.3 The individual department manager/supervisor/organization is responsible for providing PPE, test equipment and other items required for compliance with this document. The VEB reserves the right to suspend certification of Qualified Persons for cause. If a person has not actively worked as a green card or blue card holder and they are not comfortable performing green/blue card activities, it is their responsibility to re-take the initial training. If employment is terminated (with the exception of transitioning from a temporary employee to JJVC), certification is revoked. Individuals who meet the definition of contractor in **SGOP-0210** cannot be blue or green card certified.

3.2 Training for Qualified Persons:

3.2.1 All employees, temporary employees, and/or contractors whose work, or the work of those they supervise has the potential for exposure to electrical hazards, must be trained and be considered by management to be Qualified for the specific electrical hazards and tasks involved before they are exposed to such hazards.

3.2.2 This training is to ensure employees know how to recognize electrical hazards and how to protect themselves from the hazards. This certification does not imply that any Qualified Person is an electrician, master electrician or journeyman.

3.2.3 All training requirements are listed in the most current version of the Environmental, Health, and Safety Training Matrix Form Control No.**VWA-0372.** Documentation shall be generated that each employee has received all training required by the Electrical Safety Program (ESP). This documentation shall be made when the employee demonstrates proficiency in the work practices involved and all training documents shall be maintained for the duration of employment.

3.2.4 Documentation shall contain the content of the training, each employee's name, and dates of the training.

3.2.5 The pre-requisites for Qualified Person certification is to read the **SGOP-0022** and all associated documents and complete the Electrical Orientation including NFPA 70E training and relevant incidents and/or issues training.

3.2.6 Training received shall be specific to the work methods and the electrical hazards present.

3.2.7 Qualified Persons shall be trained in, and be familiar with, the skills and techniques necessary to:

3.2.7.1 Determine potential safety hazards from work on or near electrical conductors, equipment, and circuit parts, proper Safe Work Practices to be used.
3.2.7.2 Distinguish energized parts from other components of electric circuits.
3.2.7.3 Proper selection and use of electrical test equipment and PPE.
3.2.7.4 Determine the nominal voltage.
3.2.7.5 Determine the electric arc flash hazard.
3.2.7.6 Determine the safe approach distances.
3.2.7.7 Determine the proper use of special precautionary techniques to avoid or eliminate hazards such as personal protective equipment, insulating and shielding materials and insulated tools.
3.2.7.8 Conduct the decision-making process necessary to:
- Perform the job safety planning.
- Identify electrical hazards.
- Assess the associated risk

Select the appropriate risk control methods from the hierarchy of controls.
(Elimination, Substitution, Engineering controls, awareness, administrative controls and PPE) identified in the Risk Assessment.

3.2.7.9 Perform emergency procedures including reporting and proper methods of releasing victims from contact with energized conductors.
3.2.7.10 Determine the proper application and use of electrical test equipment.
3.2.7.11 Qualified Personnel shall be familiar with the operation of the equipment, required training (JSA/HECP) and comfortable with the task for the equipment they work on/near (i.e. switchgear, control side vs MCC & large breakers).
3.2.7.12 NFPA 70E – current version
3.2.7.13 Relevant incidents and/or issues.
3.2.7.14 Site-specific electrical hazards (Special Operating Conditions).

3.2.8 Hands on Skills Demonstration.
- 3.2.8.1 Distinguish energized parts from other parts.
- 3.2.8.2 Determine the nominal voltage.
- 3.2.8.3 Determine the Arc Flash Hazards.
- 3.2.8.4 Determine the proper PPE needed.
- 3.2.8.5 Determine the proper Safe Work Practices to be used.
- 3.2.8.6 Determine the proper safe approach distances.
- 3.2.8.7 The proper use of electrical test equipment.
- 3.2.8.8 Establishment of an Electrically Safe Work Condition.

3.3 Electrical Certification program for JJVC® employees and directly supervised temporary employees.

3.3.1 Candidate Selection

3.3.1.1 Blue Card holder.

A potential Blue Card holder candidate must be sponsored and approved by a member of the JJVC Electrical Board. At a minimum, the following are requirements that must be met in order to be considered for the Blue Card Certification process:

- Have 5 years or more of experience related to designing, installing and servicing electrical equipment.
- Must be a Journeyman Electrician, Master Electrician, Electrical Engineer or other highly trained and experienced employee.

3.3.2 Initial Certification

3.3.2.1 Green Card Holder.

Prior to performing electrical work, all Green Card Holders must certify by passing the following sections.

- Electrical safety related written questions test, passing score of 100%.
- General electrical written questions test, passing score of 90% or higher.
- Hands-on skills demonstration successfully completed.
- The sections that are not scored high enough may be tested again without having to retake the entire test. The person must wait 30 days prior to retake any part failed. If the Electrical Certification test is failed on the second attempt, certification is not permitted for 9 months. Proof of further skill enhancement must be presented before the entire Electrical Certification test can be given again.

3.3.2.2 Blue Card Holder.

Prior to performing electrical work, all Blue Card holders must certify by passing the following sections:

- Complete the pre-requisite items if not already completed for the Green Card holder certification.
- Obtain Blue Card Sponsor Form signed by VEB member.
- Electrical safety related written questions test, passing score of 100%.
- General electrical written questions test, passing score of 90%
- Advanced electrical written questions test, passing score of 90%.
- Hands-on skills demonstration successfully completed.
- If the electrical certification test is failed on the second attempt, certification is not permitted for 9 months. Proof of further skill enhancement must be presented before the entire Electrical Certification test can be given again.

3.3.3. Annual Recertification

3.3.3.1 Green Card Holder

- Training consisting of a 70E PowerPoint approved by the VEB and delivered through Compliance Wire including electrical safety related written questions test, passing score of 100%.
- CBT of general electrical written questions test, passing score of 90%.
- Read the **SGOP-0022**
- CBT of relevant incidents and/or issues.
- Demonstrate safety related work practices including job risk analysis utilizing Form Control No. **VCF-2745**.

Reference Form Control No. **VWA-1104** JJVC Blue/Green Card Authorization Board Certification Documentation, the following is applicable:

- Hands-on check of the FLUKE T5/Metrix approved meter.
- Inspection of personal protection equipment,'
- Inspection of VOM, amp probe or other test equipment.

3.3.3.2 Blue Card Holder

- Training consisting of a 70E PowerPoint approved by the VEB and delivered through Compliance Wire including electrical safety related written questions test, passing score of 100%.
- CBT of general electrical written questions test, passing score of 90%.
- Read the **SGOP-0022**
- CBT of relevant incidents and/or issues.
- Demonstrate safety related work practices including job risk analysis utilizing Form Control No. **VCF-2745**.

Reference Form Control No. **VWA-1104** JJVC Blue/Green Card Authorization Board Certification Documentation, the following is applicable:

- Hands-on check of the FLUKE T5/Metrix approved meter.
- Inspection of personal protection equipment,'
- Inspection of VOM, amp probe or other test equipment.

3.3.4 Electrical Work Allowed to be Performed by Green/Blue Card Holders.

3.3.4.1 General.

Energized Electrical Diagnostic Work performed within the Limited Approach Boundary of energized electrical conductors or circuit parts by Qualified Persons related to tasks such as testing, diagnostic work, voltage measuring, etc. shall be permitted to be performed without an energized electrical work permit, provided appropriate safe work practices and PPE are utilized.

Conductive articles of jewelry and clothing (such as watchbands, earrings, bracelets, rings, key chains, necklaces, metalized aprons, cloth with conductive tread, metal headgear or metal frame glasses) shall not be worn where they present an electrical contact hazard with exposed energized electrical conductors or circuit parts regardless of the voltage present.
An Unqualified Person when advised of the possible hazards, wearing appropriate PPE, continuously escorted and supervised by a Qualified Person Blue Card holder shall be allowed to cross the limited approach boundary and/or the arc flash boundary. Under no circumstance should an Unqualified Person cross the restricted approach boundary.

For all electrical repair and diagnostic work, the work alone policy is in effect.

Changing a light bulb is not considered energized electrical repair work.

3.3.4.2 Green Card Holder
Without Blue Card holder supervision or inspection prior to restoring power:

NOTE: These tasks do not require Blue Card holder inspection before power is restored.

- Diagnostic work, meter readings, inspection, and adjustments to equipment settings when equipment is finger touch protected and is 600 Volts or less.
- Work-on circuits and components less than 50 Volts AC/DC.
- Replacement of the following equipment with parts identical to the original (identical manufacturer, type, part number, etc.).
    - Lamp Changes (plug or pin type)
    - Ballast changes (plug-in type only)
    - Servo Motors (with quick disconnects)
    - Any 2 or 3 prong plug equipment covered under the single-source HECP which is known to be in good working order
    - Replace components on 600 Volts or less as long as individual wires are not required to be disconnected and reconnected, for example components having modular plugs but not the wiring inside the modular plug.
- The Green Card holder will save the failed components and keep them available for inspection by the Blue Card holder.

- Under the **direct supervision** of Blue Card holder:

- Replace electrical equipment 50 volts AC/DC and above, such as motors, fuses, transformers, starters, breakers, switches, etc.
- Assist in making modifications to new/existing wiring or components to enhance skills in a supervised environment.
- Replace components that fail soon after replacement (including those less than 50 Volts AC/DC).
- Replace failed components that have any signs of melting, burning, or any other heat related fault, regardless of voltage.

3.3.4.3 Blue Card Holder
- Diagnostic work, inspecting and having access to electrical components where it is not possible to provide finger touch protection such as disconnects, motors and transformers.
- Replace electrical equipment 50 volts AC/DC and above, such as motors, fuses, transformers, starters, breakers, switches, etc
- Modifications to existing wiring or components or add new wiring or components to existing circuits 600 Volts or less.
- Replace parts that are not identical to the original (identical manufacturer, type, part number, etc.) electrical component replacement
- Assess of the impact of using an alternative component
- As appropriate, perform testing to verify continued safe operation of the affected circuits
- For components that have any sign of melting, burning, or any heat related fault, a Blue Card holder must inspect the circuit prior to restoring power.
- Provide direct supervision to Green Card Holders when performing work required under the direct supervision of a Blue Card Holder. Before power is restored, it is the responsibility of the Blue Card holder to inspect and insure that correct wiring and proper replacement parts have been used.

3.3.5 Compliance with Electrical Safety Policy Requirements

3.3.5.1. All Blue and Green Card certified employees shall comply with the Electrical Safety Policy requirements and perform work consistent with their training and authorization.

3.3.5.2 Any Blue or Green Card Certified employee who demonstrates a willful violation of any of the requirements of the **SGOP-0022** will immediately have their badge and certification revoked. They will have to wait a minimum of 30 days, provide proof of further skill enhancement and re-certify as if they were never certified.

3.3.6 Medical First Responders

3.3.6.1 Emergency Response/Contact Release Training
Employees exposed to shock hazards shall be trained in methods of safe release of victims from contact with exposed energized electrical conductors or circuit parts. Refresher training shall occur JJV Occupational Health.

3.3.6.2 First Aid, Emergency Response, and Resuscitation
Employees responsible for responding to medical emergencies shall be trained in:
- First aid and emergency procedures.
- cardiopulmonary resuscitation (CPR) - refresher training shall occur annually
- The use of an automated external defibrillator (AED) if an employer's emergency response plan includes the use of this device - refresher training shall occur JJV Occupational Health.

Employers shall verify at least annually that required emergency procedures training is current and is properly documented.

3.3.7 **Unqualified Persons Training Requirements**:
Prior to work and every two years, Unqualified Persons must be trained in the following:
- Hazards associated with electricity.
- Effects of electricity on the body.
- Electrical safety-related work practices required preventing electrical shock and/or other electrically related injuries.

Training requirements are listed in most current version of the Environmental, Health, and Safety Training Matrix Form Control No. **VWA-0372.**

3.3.8 Contractor Companies:

3.3.8.1 The previous level of electrical safety training for contractors, outside service, and other non-JJVC personnel shall be evaluated before they are brought on site to determine whether or not this training is acceptable for the work to be done using Form Control No. **VCF-2337,** *Electrical Safety Related Work Practices Pre-Approval Evaluation Checklist*. Once completed by the contractor, the checklist shall be evaluated by the JJVC Contractor Representative in coordination with the **SGOP-0210** requirements. This evaluation shall be conducted periodically at a frequency as determined by the work and in no case longer than once per year. Complete **VCF-2337** review annually.

3.3.8.2 Any contractor company personnel (**SGOP-0210**) performing electrical work that could expose to them to electrical hazards shall receive:
- **SGOP-0022.**
- Hazardous Energy Control Procedure specific training.
- Arc Flash requirements.

3.3.8.3 This type of work shall be performed only by a journeyman or master electrician.

3.3.8.4 If the contractor company personnel are not a journeyman or master electrician, then No.**VCF-2337** shall be approved by the JJVC Contractor Representative in coordination with all the **SGOP-0210** requirements.

3.3.8.5 Form Control No.**VCF-2337** is required for contractor company personnel hired to perform a specific task who has completed appropriate training to safely service specific devices, machinery or equipment and has been briefed on local site practices, signage and requirements as listed above – not to perform master electrician or journeyman type work (electrical installations, etc.).

3.3.8.6 JJVC employees shall report any observed violations to the contract employer immediately. They shall then notify their supervisor who shall determine and take appropriate action to remove the hazard and correct the violation.

3.3.8.7 Contractors, outside service, and other non- JJVC personnel shall inform JJVC of:

- any unique hazards presented by the work.
- any unanticipated hazards found during the work execution not already communicated.
- measures taken by the contractor to correct violations identified by JJVC.

3.3.8.8 There shall be a documented meeting using Form Control No. **VCF-2222** between the JJVC representative and the contract employer before beginning work.

4.0 Safe Work Procedure Requirements

4.1 All electrical work (diagnostic or repair) 50 volts AC/DC and above requires an attendant.

4.2 Voltage rated and insulated tools, test equipment, and handling equipment shall be used when working inside the limited approach where contact either accidental or intentional contact is made. All other items inside the limited approach boundary must be non-conductive.

4.3 Sufficient illumination shall be provided to ensure Qualified Persons can avoid exposed energized components. Proper illumination levels shall be consistent with J&J standards.

4.4 Employees shall not work within the Limited Approach Boundary of energized electrical conductors or circuit parts where an electrical hazard exists while their alertness is recognizably impaired due to illness, fatigue, or other reasons.

4.5 Employees shall be alert to changes in job scope that may result in different or additional hazards.

4.6 Alerting methods shall be used for work involving an installation of multiple pieces of similar or identical equipment, sometimes referred to as “look alike” equipment. These alerting methods shall be effective in identifying the electrical hazards, and in establishing proper awareness among the Qualified Persons performing the work, relating to being sure the correct piece of equipment is addressed.

4.7 Employees are not allowed to reach blindly into areas that might contain exposed energized electrical conductors or circuit parts where electrical hazards exists.

4.8 If a piece of equipment is in an unsafe condition (i.e. melted wires, overheating, smoke, etc.) the equipment shall be de-energized, locked and tagged, and the unsafe condition corrected.

4.9 Single line drawings shall be maintained to accurately represent field conditions.

4.10 Overcurrent protection of circuits and conductors shall not be modified.

4.11 The cause of a protective device operation due to a fault condition (i.e. short circuit, phase to phase fault, phase to ground fault, etc.) shall be identified and corrected by a Qualified Person before a circuit is re-energized. Where it is determined that the protective device operation was caused by an overload rather than a fault condition, examination of the circuit shall not be required before the circuit is reenergized.

4.12 Energized components operating at 50 volts AC/DC and below need not be de-energized as long as there is no exposure to electrical burns or to explosion due to electric arcs.

4.13 When arc flash energy exceeds 40 cal/cm^2 at the normal working distance, and the work to be done is such that arc flash PPE is required, the equipment shall be de-energized before working within the Arc Flash Boundary of the electrical conductors or circuit parts.

NOTE: Work may be performed, however, on such equipment if methods are used to reduce the arc flash hazard, such as increase the working distance or decrease the fault clearing time, such that the employee would not be exposed to arc flash hazards exceeding 40 cal/cm^2 should an arc flash event occur. Examples of these methods would include the use of remote racking devices, remote operating devices, cable operated circuit breaker operators, circuit test points wired to a remote terminal strip, enabling instantaneous trip functions during maintenance, etc.

4.14 No individual on JJVC premises (including JJVC/J&J associate, temporary employees and contractors) shall be permitted to work (diagnostic or repair) in an area where they are likely to encounter electrical hazards unless they are "Qualified Personnel".

4.15 JJVC/J&J associates and temporary employees that have been certified as a Qualified Person can be identified with an electrical safety training card (Green or Blue card).

4.16 Contractors are qualified by their companies and are either journeyman electricians, master electricians or have Form Control No. **VCF-2337** completed and approved by a JJVC VEB member.

4.17 If energized electrical repair work is needed, only approved contractors shall perform the work, and only after an Live Electrical Work Permit in the ePTW system has been issued and approved.

4.18 JJVC associates are not allowed to work on High Voltage equipment.

4.19 To facilitate the establishment of an electrically safe work condition and minimize Arc Flash exposure, the following shall be adhered:
- All new installations greater than 120 VAC must have an arc flash calculation
- All user serviceable equipment must be calculated 1.2 cal/cm^2 or less at 18" ~~or 24"~~ (working distance)
- Adequate probe holes in finger touch guarding shall be available for testing/troubleshooting/verification
- All new disconnects shall be finger touch protected.

5.0 Establishment of an Electrically Safe Work Condition

5.1 Apply Lock and Tag in accordance with the applicable HECP (refer to **SGOP-0028**) Use the Left (Off) Hand Rule when operating disconnects or breakers.

5.2 One of the following methods shall be used:

5.2.1 A) Verification using an approved permanently installed Absence of Voltage Tester (AVT) per **VWA-0835.**

B) Use a FLUKE T5 or Metrix MX 57EX (intrinsically safe areas only) Tester.
- Check the meter is operating correctly on a known voltage source.
- Verify the circuit de-energized by checking each conductor both phase-to-phase and phase-to-ground.
- Re-Check the meter is operating correctly on a known voltage source.

5.2.2 Electrically Safe Work condition exists only if 5.1 and either step 5.2.1 A) or B) of the above are followed and the line side of the main disconnecting device is Finger Touch Protected such that the risk of accidental contact is eliminated and there are no other energized components present. No electrical PPE is required and conductive tools are permitted to be used.

5.2.3 The Electrically Safe Work condition shall be re-established if power is applied.

5.2.4 The circuit shall be considered energized until it is placed in an Electrically Safe Work Condition. Use of control devices or interlock circuit is not an acceptable method of lockout/tagout.

6.0 Energized Electrical Repair Work

6.1 No JJVC® employee or contractor may perform Energized Electrical Repair Work on "energized circuits" 50 volts AC/DC and above.

6.2 If the primary side of a main disconnect switch and/or the main distribution busses require work, only licensed electricians who are also considered “Qualified” may perform this task.

6.3 All Energized Electrical Repair Work shall be performed by “Qualified Personnel”, those who are allowed to work on electrical devices, up to 600 volts.

6.4 Such work includes, but not limited to, tighten, disconnect, adjust or connect wires and terminals.

6.5 Routine operations of equipment that is intended by the manufacturer to be done while energized, when performed by a Qualified Person wearing appropriate PPE and using proper safe work practices is not considered work that requires an Energized Electrical Work Permit. Examples include removing draw-out MCC buckets that are guarded and shuttered, with the main switch in the open position such that the operation is not done under load, plugging in or unplugging starters, circuit breakers, disconnects, etc. provided the operation is designed and intended by the equipment manufacturer.

7.0 Energized Electrical Diagnostic Work

7.1 Only Qualified Personnel may perform diagnostic work on or near energized parts operating at 50 volts AC/DC or more.

7.2 The Qualified Person must understand the Arc Flash Labels information.

7.3 The following steps must take place prior to opening an electrical cabinet door:

- 7.3.1 Identify the disconnecting means of all electrical sources to the panel.
- 7.3.2 Identify the limited, restricted shock and arc flash protection boundary and PPE level for each source.
- 7.3.3 Ensure that all Unqualified Personnel are outside the arc flash protection and the limited approach boundaries.
- 7.3.4 Inspect the required PPE.
- 7.3.5 Inspect the required equipment.

7.4 Prior to perform Diagnostic Work, once the door of the electrical cabinet is open, the Qualified Person must verify that all electrical components above 50 volts AC/DC are “Finger Touch” protected.

- 7.4.1 In the event that the finger touch protection is missing, the door must be closed and the diagnostic work may only be performed after all guards have been properly re-installed while the electrical source(s) are locked and tagged out.
- 7.4.2 Only where NOT POSSIBLE (must be approved through electrical engineering assessment) to provide “Finger Touch” protection, a Blue cardholder or Contractor company journeyman or master electrician must perform the diagnostic work as per the rules described in the “Blue cardholder classification: electrical work allowed” below. Voltage rated gloves with leather protectors shall be worn as additional PPE. Any disconnects/panels are not finger touch protected shall be externally labeled.

7.5 **SGOP-0028** shall be followed when bypassing electrical safety interlocks.

7.6 If electrical conductors and/or circuit parts are not placed in an Electrically Safe Work Condition, the work involved shall be considered energized electrical work and shall be performed by written permit only.

7.7 Exceptions: (energized electrical work not requiring a permit)
- Normal testing and diagnostic work activities such as voltage measurements and diagnostic tasks on electrical equipment performed by Qualified Persons using appropriate safe electrical work practices and electrical protective clothing and equipment.

7.8 Only a Qualified Person shall work on or near exposed energized conductors or circuit parts. In areas where only Qualified Persons are allowed, warning signs, pylons, barricades, etc. shall be used where necessary to warn Unqualified Persons about the potential for injury due electrical hazards, and to prohibit Unqualified Persons from entering the area. Barricades shall be placed no closer than the Limited Approach Boundary or the Arc Flash Boundary, whichever is greater.

7.9 An attendant who has been briefed on the hazards shall be posted at all entrances to warn and prevent entry by Unqualified Persons. The briefing shall include the following at a minimum:
- How to summons emergency help
- Release methods if electrical shock occurs
- The attendant shall remain in the area as long as there is a potential for employees to be exposed to the electrical hazards.
- The attendant shall not be distracted.

7.10 If an unqualified person is the attendant, the brief shall include the above and the following:
- Not to cross the most limiting boundary.
- Not to allow anyone, other than those performing work, in the area.

7.11 During normal operation, electrical equipment enclosure covers and doors shall be in place as intended by the manufacturer with all fasteners secured. Unqualified Persons are prohibited from opening an electrical equipment enclosure for any reason. If an unattended open enclosure is observed, this condition shall be reported to a Qualified Person immediately.

8.0 Job Safety Planning and Job Briefing

8.1 Job Hazard Analysis (Risk Assessment)
Prior to beginning any electrical work, a Job Hazards Analysis shall be performed for each activity that has the potential for personnel to be exposed to injury due to electrical hazards, for work involving both AC and DC systems. The analysis shall include methods to be used by the employee before work is started to:

8.1.1 Review the description of the job and the individual tasks.
8.1.2 Identify hazards.
8.1.3 Assess risks, including human error.
8.1.4 Implement appropriate risk control(s) according to the following hierarchy of controls in the order listed:
- Elimination
- Substitution
- Engineering controls
- Awareness
- Administrative controls
- PPE

8.2 A Job Hazards Analysis (Risk Assessment) shall be completed by a Qualified Person and include at a minimum the following:

8.2.1 Isolation means (fed from)
8.2.2 Determine Arc Flash Boundary
8.2.3 Determine Arc Flash Energy level at the working distance (18 inches)
8.2.4 Determine and don proper PPE (Arc Flash and shock protection)
8.2.5 Determine nominal voltage
- Limited Approach Boundary
- Restricted Approach Boundary

8.2.6 Review for Special Operating Conditions
8.2.7 Description of the safe work practices to be used
- Station Attendant (Brief as required)
- Other means to restrict access (as needed)
- Insulated/Voltage Rated tools and test equipment as required
- Non-conductive lighting (as needed)

8.2.8 Apply adequately rated personal protective ground sets (as needed)
8.2.9 Live Electrical Work Permit in the ePTW system (as needed)

8.3 The Job Hazards Analysis (Risk Assessment) results shall cover the description of job and the individual tasks and shall be communicated to all who could be exposed to the hazards prior to beginning work. This is required as a pre-job task and is not required to be formally documented.

8.4 When performing diagnostic work on energized circuits, work practices appropriate for the hazards involved (i.e. use of insulated tools, appropriate personal protective equipment, use of barriers, etc.) shall be used to protect employees from contact with energized parts directly with any part of their body, or indirectly through some other conductive object.

8.5 Personal Protective Equipment:
Consult Form Control No.**VWA-0836,** for proper selection, care, handling, maintenance, inspection, testing and use of electrical PPE and Other Electrical Protective Equipment. Consult Form Control No.**VWA-0834** for the purchasing specifications of PPE and Other Electrical Protective Equipment. Refer to the arc flash label to identify personal protective equipment needed for protection against electrical hazards.

8.6 Clothing shall cover potentially exposed areas as completely as possible. Shirt sleeves shall be fastened at the wrists, and shirts and jackets shall be closed at the neck.

8.7 Reference Form Control No.**VWA-0839** for anything not having an arc flash hazard label.

8.8 Proper care, use and inspection of the PPE including lab coats, long sleeve shirts, hairnets and/or beard nets are the responsibility of the Qualified Persons.

8.9 PPE is required to be inspected prior to each use. Reference Form Control No.**VWA-0836** for inspection and testing frequencies of electrical PPE.

8.10 JJVC® employees responsible for work being performed must ensure that outside contractor/vendor personnel comply with the JJVC® Electrical Safety Policy and **SGOP-0210.**

9.0 Special Operating Conditions

9.1 Special Operating Conditions are those areas, installation methods, equipment, work practices, etc. unique to the location that have the potential to present safety hazards to personnel, that would not normally be encountered at other similar facilities.

9.2 Special operating conditions include electrically classified hazardous areas including the Isopropyl Alcohol and chemical storage areas, wet locations (sterilizers), diagnostic work IS barriers, waste plastic/resin areas, and areas with special lighting.

9.3 Where flammable atmospheres could possibly be present, only explosion-proof or intrinsically safe electrical equipment shall be used that is in compliance with the appropriate codes and regulations regarding classified areas.

9.4 Appropriate safe approach distances for Qualified and Unqualified Personnel shall be maintained while there is a risk of exposure to electrical hazards.

9.5 Some distribution panels have feeds from emergency generators and/or UPS systems. Ensure electrical isolation of all sources prior to work.

9.6 Any person working in a congested space containing exposed energized components shall use protective shields, barriers, or insulating materials as necessary to avoid inadvertent contact.

9.7 Doors, hinged panels, and other covers shall be secured to prevent their swinging into an individual and causing contact to exposed energized components.

9.8 Ladders are not allowed for any type of electrical work. Non-conductive platforms/lifts with standard rails are allowed. The only exception is to verify isolation of equipment after the equipment has been isolated utilizing the appropriate Hazardous Energy Control Plan.

9.9 Battery Rooms/UPS Backup Systems that exceed 50 volts AC/DC

9.9.1 Limit access to battery compartment and bus terminals to Blue Card or Qualified electrician (Journeymen or above).

9.9.2 All systems >50 volts AC/DC need to be labeled with voltage and arc-flash rating based on the short circuit current rating. If no short circuit current rating is available, then HRC 4 PPE is required. If short circuit current rating is less than 1KA HRC 0 PPE is required.

9.9.3 If during maintenance/inspection a battery has been found to be damaged, then you are required to follow clean up procedures and wear appropriate PPE for handling the electrolyte.

9.9.4 Tools and equipment for working on batteries shall be insulated for at least the maximum working voltage.

NOTE: Ref NFPA 70E 2021 edition Article 320

9.10 Grounding and Bonding of Fixed Equipment in Electrically Unclassified Areas

9.10.1 Grounding of fixed equipment shall be maintained to ensure electrical continuity. The current revision of **SGOP-0041** shall be referenced for preventative maintenance requirements in electrically classified areas.

9.10.2 Grounding and bonding of fixed equipment that personnel can come in contact with in electrically unclassified areas shall be documented and, at a minimum, meet the following:

- Initial Installation: Continuity test (10 ohms or less to building ground) and proper torque verification (hand tool snug)
- Annual visual inspection: Visually inspect for damage (corrosion, discoloration, heat damage, etc.,) and that all grounds are physically attached.
- Mechanical Inspection (other than MCC and Distribution Panels): Every three years (unless documented manufacturer or NFPA requirements state different frequency) continuity test (10 ohms or less to building ground) and proper torque verification (hand tool snug).

- Mechanical Inspection – MCC and Distribution Panels: Every five years (unless documented manufacturer or NFPA requirements state different frequency) continuity test (10 ohms or less to building ground) and proper torque verification (hand tool snug).

9.11 Overhead Power Lines

9.11.1 If work is to be performed near overhead power lines that could present an electrical hazard, the lines shall be de-energized and grounded, or other safe work practices shall be used based on a Job Hazards Analysis (Risk Assessment) before work begins.

9.11.2 A Qualified Person shall determine if overhead lines are insulated for the operating voltage, covered with a material not considered acceptable as insulation for the operating voltage, or are bare uncovered and un-insulated. Lines that are covered with a material not considered acceptable as insulation for the operating voltage shall be considered bare, un-insulated conductors, and appropriate protective measures shall be taken.

9.11.3 Appropriate safe approach distances for Qualified and Unqualified Personnel shall be maintained while there is a risk of exposure to electrical hazards.

9.12 Cutting or Drilling

9.12.1 Before cutting or drilling into equipment, floors, walls, or structural elements where a likelihood of contacting energized electrical lines or parts exists, a Job Hazards Analysis (Risk Assessment) shall be performed to:

- Identify and mark the location of conductors, cables, raceways, or equipment
- Create an electrically safe work condition
- Identify safe work practices and PPE to be used

10.0 ARC Flash Labeling

10.1 Electrical equipment such as switchboards, panel boards, industrial control panels, meter socket enclosures, and motor control centers that are likely to require examination, adjustment, servicing, or maintenance while energized shall be field-marked with an electrical hazards warning label containing all the following information:

10.1.1 - Highest voltage present
10.1.2 - Arc Flash Boundary
10.1.3 - At least one of the following
- Available incident energy and the corresponding working distance, or the Arc Flash PPE Category for the equipment.
- Minimum arc rating of clothing
- Site-specific level of arc rated PPE.

10.1.4 The method of calculating, and the data to support the information for the label shall be documented. Where a review of the arc flash hazard risk assessment identifies a change that renders the label inaccurate, the label shall be updated.

10.1.5 The owner of the electrical equipment shall be responsible for the documentation, installation, and maintenance of the filed-marked label. The label shall be in accordance with Form Control No. **VWA-0839**.

10.1.6 All equipment powered by 3-phase power must have an Arc Flash Label at the primary means of lockout for that equipment. Any sub panel, which is listed in the machine specific HECP, should also have an Arc Flash label. If it is a power source that requires an arc flash label (i.e. 480 volt ac or any 3-phase power) and is used for the local lock and tag then it requires an arc flash label to be placed on it. A junction box only requires a label if it is routinely accessed. A VEB member shall approve addition of new, or removal of any existing arc flash label.

10.1.7 Each area shall maintain an inventory of equipment requiring arc flash labels. The inventory at a minimum shall include the name of the equipment, the fed from source, voltage and the associated calculation or single-phase memo reference. Equipment requiring Arc Flash Labels shall be inventoried annually including visual observation of the label to ensure it is still present and legible and verification of the referenced calculation on the inventory. A quarterly inspection of all labeling shall be conducted using Form Control No.**VCF-1774**.

10.1.8 The arc flash hazards analysis shall be reviewed and revised, or reaffirmed as still applicable and correct, as required whenever:

- the internal electrical power distribution system is modified
- there are changes in the available fault current from the outside utility source,
- codes, standards or policies related to determining arc flash energy change, or
- at a frequency not to exceed five years, whichever is sooner

NOTE: The current revision of SGOP-0318 covers these modifications/renovations. The Facilities Planning and Engineering VEB member is responsible for the documented review. Each area VEB member is responsible for delivering information that may impact the arc flash hazard analysis for their respective area of responsibility.

10.1.9 Unless an Arc Flash Hazards Analysis has been performed and an Arc Flash Label is attached to the equipment, Form Control No.**VWA-0839** shall be referenced for proper PPE.

11.0 Portable Electrical Cord & Plug Equipment

11.1 The following requirements are for the use of cord and plug connected equipment and extension cord sets. Employees shall not be permitted to use any equipment that has not met the following requirements. Plugs and cords shall be installed according to the National Electrical Code.

11.2 Reference Form Control No.**VWA-0837** for proper selection, use, care, maintenance, and testing for cord and plug equipment. Reference Form Control No. **VWA-0833** for purchasing specifications for electrical plug and cord connected equipment and tools.

11.3 Before each use, portable electrically operated cord and plug connected equipment and tools shall be visually inspected for external defects. If there is a defect or evidence of damage that might expose someone to injury, the defective item shall be taken out of service until it is repaired or replaced.

11.4 Plugs and cords shall not be used:

11.2.1 As a substitute for fixed wiring of a structure.
11.2.2 As a run/concealed through walls, ceilings, or floors.
11.2.3 As permanent wiring run through doors, windows, or similar openings.
11.2.4 Where attached to a building surface.
11.2.5 The use of household or office type cords/power strips is not permitted in industrial environments.
11.2.6 “Daisy-chaining” multiple outlet extension cords (computer strips, etc.)

11.5 Portable equipment shall be used in accordance with the manufacturer’s instructions and safety warnings.

11.6 Grounding-type equipment requirements.

11.6.1 A flexible cord used with grounding-type equipment and extension cord sets shall contain an equipment-grounding conductor. The following are the requirements when using such equipment.

- Attachment plugs and receptacles may not be connected or altered in a manner that would prevent proper continuity of the equipment grounding conductor
- Adapters that interrupt the continuity of the equipment grounding connection shall not be used.

11.7 Care of portable electrical/electronic equipment and electrical cord sets.

11.7.1 Handling

- Portable equipment shall be handled in a manner, which will not cause damage.
- Flexible cords connected to equipment shall not be used for raising or lowering the equipment.
- Flexible cords may not be fastened with staples or otherwise hung in such a fashion as could damage the outer jacket of insulation.

11.8 Where an attachment plug is to be connected to a receptacle, the relationship of the plug and receptacle contacts shall first be checked to ensure that they are of proper mating configurations.

11.9 Strain relief devices are required to protect from internal cord and conductor damage due to flexing at the exit point from the rigid hardware or electrical fitting on the equipment.

NOTE: Stationary plug and cord connected equipment (such as personal computers, water coolers, emergency lights, household appliances, and other such machinery), that are installed such that the connecting cord and plug are not subject to physical damage during normal use, need not be inspected or tested except when they are initially installed, relocated or repaired. Reference the Form Control No. VWA-0837.

11.10 Ground fault protection for personnel shall be provided for all maintenance and construction activities when using cord and plug connected tools and equipment. This ground fault protection for personnel shall be provided by either using a listed ground fault circuit interrupter (GFCI) device or by the implementation of assured equipment grounding conductor program.

11.11 GFCI device shall be plugged directly into the electrical outlet followed by the extension cord (if used) and then by the portable electric tool. GFCI devices must be rated for the connected load and periodically tested as per manufacturer's instructions.

11.12 All flexible extension cords shall contain a grounding conductor. No more than 100 feet of extension cord, supplying rated current for the wire size - minimum 14-gauge (as marked on the cord by the manufacturer) shall be used on site.

12.0 Portable Electrical and Electronic Test Equipment

12.1 The requirements for the proper selection, use, care; inspection, maintenance and testing for Electrical and Electronic Test Equipment are established in Form Control No.**VWA-0838.**

12.2 Consult Form Control No.**VWA-0835** for purchasing specifications for Electrical and Electronic Test Equipment.

12.3 Portable electrical and electronic test equipment shall be periodically inspected and tested by a Qualified Person at least annually in Form Control No.**VWA-1104**.

12.4 Only Qualified Persons shall perform testing work on electric circuits or equipment that is rated at 50 volts AC/DC or more.

13.0 Conductive or Wet Work Locations

13.1 Battery or air driven equipment shall be the first choice to use in conductive or wet work locations. If this type of equipment is not available, portable electric equipment and flexible cords used in conductive work locations, such as those inundated with water or other conductive liquids, or in job locations where persons are likely to contact water of conductive liquids shall be approved for use in those locations.

13.2 Areas containing water hoses, near water dispensing areas or sinks, areas designated as wet areas, water borne equipment (examples include water baths, ultrasonic sinks, water circulators, etc.) and exterior locations on the roof or outside grounds should have all general-purpose utility outlets (120 or 240 VAC) protected by Ground Fault Circuit Interrupters (GFCI).

14.0 Connecting Attachment Plugs Requirement

14.1 The following are the requirements for connecting attachment plugs.

14.1.1 An individual's hands shall not be wet when plugging and unplugging flexible cords and cord and plug-connected equipment.

14.1.2 Energized plug and receptacle connections shall be handled only with insulating protective equipment if the condition of the connection could provide a conducting path if the person's hands (for example, a cord connector is wet).

NOTE: If it is not possible to remove the potential for shock through the use of insulating protective equipment, the energized circuits must be de-energized at the source before plugging or unplugging the apparatus.

14.2 Locking-type connectors shall be properly secured after connection.

15.0 Power Distribution Systems Maintenance

15.1 Power distribution systems shall be maintained in accordance with local, state and federal requirements (including NPFA 70B) and manufacturer's instructions to prevent failures, explosions, or fires as a result of improperly maintained, services or adjusted equipment - only blue card associates approved by the JJVC® Electrical Board shall work on this type of equipment.

16.0 Excavation Work Done Near Underground Electrical Conductors

16.1 An Electrical Engineer shall review and approve any excavation work done near underground electrical conductors including consultation with outside professional services as required. Facilities Planning maintains facility drawings that document the location of such conductors.

16.2 Where a reasonable possibility exists for contacting any underground electrical lines or equipment, necessary steps shall be taken to identify the location of the underground electrical lines or equipment, and a Job Hazards Analysis (Risk Assessment) shall be performed to determine appropriate safe work practices to be used during the excavation. If necessary, the circuit shall be de-energized during the time the excavation work is in progress. Excavation PTW Form shall be completed prior to beginning work.

17.0 Electrical Safety Program Auditing.

17.1 The overall electrical safety program shall be periodically audited to ensure that the principles and procedures of the program are in compliance with this procedure and applicable codes and standards. Audits shall be performed at intervals not to exceed 3 years.

17.2 The lockout/tagout program and procedures shall comply with **SGOP-0028**.

17.3 Where the audit(s) determines that the principles and procedures of the electrical safety program are not being followed, appropriate revisions to the training program or procedures shall be made.

17.4 All audits shall be documented.

18.0 Responsibilities

18.1 The following are the responsibilities of each department or individual that is involved with Electrical Safety Program.

18.1.1 JJVC® Employee
- Compliance with all requirements in this document.
- Qualified Persons shall protect others associated with work in progress and shall perform electrical tasks as per training and authorization.
- Unqualified Persons shall follow the training and/or instructions from Qualified Persons

18.1.2 Area Director/Plant Manager
- Implementing this **SGOP-0022** and ensuring employees compliance to all requirements in this document
- Ensuring compliance with all Electrical Safety requirements including the annual self-assessment.
- Providing PPE, test equipment and other items required for compliance with this document.
- Proper maintenance of the electrical equipment in accordance with manufacturer's recommendations and consensus standards, and any local, state and federal requirements.

18.1.3 Environmental, Health, and Safety (EH&S)
- To serve as technical/regulatory specialist.
- To administer the activities related to "Electrical Safety."
- Select members of the VEB.

18.1.4 JJVC® Electrical Board (Local Electrical Safety Work Practice Team)
- To sponsor and approve the Blue Card holders.
- Retain records of electrical safety certification.
- In conjunction with the EH&S, to:
  - Keep this document current with OSHA, NEC, and NFPA 70E and other applicable standards and requirements and ensure training content developed and administered for green/blue card holders of any relevant changes.
  - Identify the appropriate NFPA 70E training (material, trainer, documentation, etc) for the Qualified and Unqualified Persons.
  - Identify an internal resource to administer the scheduling and document retention for the training, certification and instrument review.
  - Electrical engineers are the designated knowledgeable individual with the authority to oversee electrical installations, activities, and methods from a safety and code compliance perspective.
  - Approve installation of new wiring from distribution panel to equipment.
  - Ensure permit is in place from City of Jacksonville Building Department, if required.

18.1.5 Electrical Engineers (JJVC's consultants and contractors)
- deliver equipment compliant to the applicable codes and standards. (Refer to the current revision of **SGOP-0023**)
- ensure appropriate permits are issued by the City of Jacksonville Building Department, if required.
- minimize hazards of electricity, including but not limited to, the following:
- incorporate any advances in technology that result in safety improvements to minimize electrical hazard potential (i.e. IEC Finger Touch Protection components, polycarbonate (Lexan®) barriers, remote racking devices, GFCI's, arc flash reduction measures, etc.).
- use the Risk Assessment process at each phase of the job or project, ensuring that choices made in the course of design, selection, installation and operation serve to eliminate risk, reduce frequency of exposure, reduce magnitude or severity of exposure, enable the ability to achieve an electrically safe work condition, and otherwise enhance the effectiveness of the safety related work practices contained in the NFPA 70E standard.

- make every effort to reduce the resulting electrical arc flash incident energies, regardless of the level, and electrical shock hazards at the equipment to the lowest possible levels, including their elimination all together.
- ensure the arc flash risk assessment is reviewed and revised, or is reaffirmed as still applicable and correct, as required. whenever:
    - the internal electrical power distribution system is modified
    - there are changes in the available fault current from the outside utility source,
    - codes, standards or policies related to determining arc flash energy change, or
    - at a frequency not to exceed five years, whichever is sooner

NOTE: The Facilities Planning and Engineering VEB member is responsible for the documented review. Each area VEB member is responsible for delivering information that may impact the arc flash hazard analysis for their respective area of responsibility. The current revision of SGOP-0318 covers these modifications.

19.0 Training

19.1 The training requirement for this document is "Read Only" for all employees in the organizations identified in the **OVERVIEW** section. This training must be conducted prior to work and refresher training given as deemed necessary by Department Directors or as a result of deficiencies identified during internal self-a assessments.